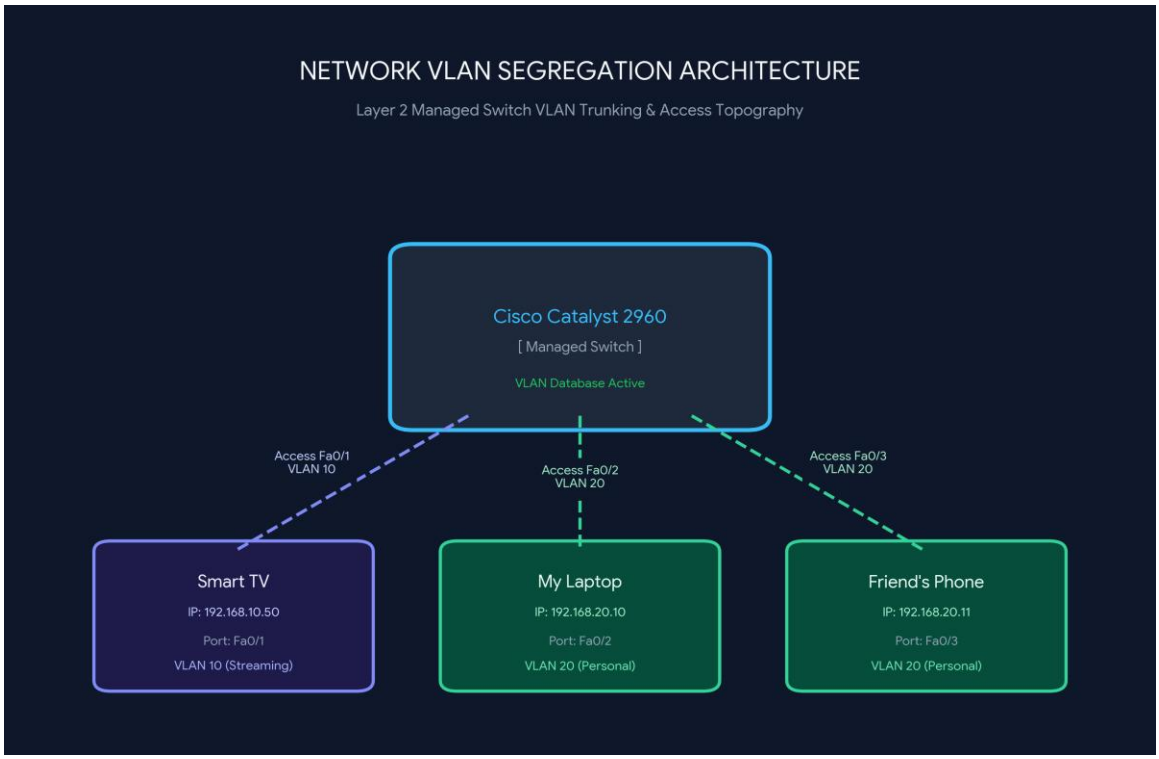
VLAN Configuration & Network Automation Report

Question 1

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

Below is the high-contrast dark theme network topology diagram illustrating three endpoint devices connected to a central Cisco Layer 2 Switch with logical VLAN separation:



Topology Description: Diagram Details & Logical Breakdown:

1. Smart TV: Connected to switch port FastEthernet0/1, assigned to VLAN 10 (Streaming Network, Subnet 192.168.10.0/24).

2. My Laptop: Connected to switch port FastEthernet0/2, assigned to VLAN 20 (Personal Devices Network, Subnet 192.168.20.0/24).

3. Friend's Phone: Connected to switch port FastEthernet0/3, assigned to VLAN 20 (Personal Devices Network, Subnet 192.168.20.0/24).

Logical Isolation: Devices in VLAN 20 (Laptop and Phone) reside in the same broadcast domain and can communicate at Layer 2. However, traffic from VLAN 10 (Smart TV) is strictly segregated at Layer 2 and cannot communicate with VLAN 20 without a Layer 3 Router/Inter-VLAN Routing.

2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.
Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

Answer & Solution:

In Cisco Packet Tracer, a Catalyst 2960 switch is deployed with 5 PCs connected to ports Fa0/1 through Fa0/5.

1. Cisco IOS CLI Configuration: Step 1: Switch VLAN Creation & Port Assignment Commands

```
Switch> enable
Switch# configure terminal

! Create VLAN 10 and assign name
Switch(config)# vlan 10
Switch(config-vlan)# name STREAMING_NET
Switch(config-vlan)# exit

! Create VLAN 20 and assign name
Switch(config)# vlan 20
Switch(config-vlan)# name PERSONAL_NET
Switch(config-vlan)# exit

! Assign Ports Fa0/1 - Fa0/3 to VLAN 10 (3 PCs)
Switch(config)# interface range fastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

! Assign Ports Fa0/4 - Fa0/5 to VLAN 20 (2 PCs)
Switch(config)# interface range fastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# end
Switch# copy running-config startup-config
```

2. Network Addressing Table: Step 2: IP Address Table

- PC-1 (Fa0/1, VLAN 10): 192.168.10.10 / 255.255.255.0
- PC-2 (Fa0/2, VLAN 10): 192.168.10.11 / 255.255.255.0
- PC-3 (Fa0/3, VLAN 10): 192.168.10.12 / 255.255.255.0
- PC-4 (Fa0/4, VLAN 20): 192.168.20.10 / 255.255.255.0
- PC-5 (Fa0/5, VLAN 20): 192.168.20.11 / 255.255.255.0

3. Connectivity Verification: Step 3: Verification & Test Results

1. Intra-VLAN Ping Test (PC-1 to PC-2, both in VLAN 10):
Command: PC-1> ping 192.168.10.11
Result: SUCCESS (4/4 Replies Received, latency <1ms).

2. Inter-VLAN Ping Test (PC-1 in VLAN 10 to PC-4 in VLAN 20):
Command: PC-1> ping 192.168.20.10
Result: FAILED (Request Timed Out). This confirms that Layer 2 isolation is actively functioning between VLAN 10 and VLAN 20.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Scenario: PC-3 is initially connected to FastEthernet0/3 in VLAN 10. We will reconfigure FastEthernet0/3 to move PC-3 into VLAN 20 and adjust its IP address accordingly.

BEFORE Migration Configuration & Testing: Initial State (BEFORE):

```
! Verify current port status on Switch
Switch# show mac address-table interface fastEthernet 0/3
Switch# show vlan id 10
! Output shows FastEthernet0/3 is member of VLAN 10

! Test ping from PC-3 (192.168.10.12) to PC-1
(192.168.10.10) in VLAN 10:
PC-3> ping 192.168.10.10
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128
(SUCCESS)

! Test ping from PC-3 (192.168.10.12) to PC-4
(192.168.20.10) in VLAN 20:
PC-3> ping 192.168.20.10
Request timed out. (FAILED)
```

Reconfiguration CLI Commands: Switch CLI Reconfiguration Commands (MOVE PROCESS):

```
Switch# configure terminal
Switch(config)# interface fastEthernet 0/3
Switch(config-if)# description Moved PC-3 to Personal
Network
```

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# end
Switch# write memory
```

AFTER Migration Configuration & Testing: Final State (AFTER):

Note: On PC-3, update the IP address configuration from 192.168.10.12 to 192.168.20.12 (Subnet 255.255.255.0) to match VLAN 20's IP subnet.

```
! Verification on Switch:
Switch# show vlan id 20
! Output confirms FastEthernet0/3 is now listed under VLAN
20

! Test ping from PC-3 (192.168.20.12) to PC-1
(192.168.10.10) in VLAN 10:
PC-3> ping 192.168.10.10
Request timed out. (FAILED - Expected due to VLAN
segregation)

! Test ping from PC-3 (192.168.20.12) to PC-4
(192.168.20.10) in VLAN 20:
PC-3> ping 192.168.20.10
Reply from 192.168.20.10: bytes=32 time=1ms TTL=128
(SUCCESS!)
```

***4*. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).**
Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Method 1 (Cisco IOS Range Command): Method 1: Cisco IOS Native 'interface range' Automation (Packet Tracer / NetSim / GNS3)

```
Switch> enable
Switch# configure terminal

! Step 1: Create VLAN 30
Switch(config)# vlan 30
Switch(config-vlan)# name GUEST_GUEST_NET
Switch(config-vlan)# exit

! Step 2: Use interface range to select 10 ports (Fa0/1 to
Fa0/10) simultaneously
Switch(config)# interface range fastEthernet 0/1 - 10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# spanning-tree portfast
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit

! Step 3: Verify configuration
Switch# show vlan id 30
```

Method 2 (Python Netmiko Loop Script): Method 2: Python Automation Script (GNS3 / Real Cisco Switches)

```
from netmiko import ConnectHandler

# Switch connection details
```

```
switch_device = {
    'device_type': 'cisco_ios',
    'host': '192.168.1.254',
    'username': 'admin',
    'password': 'Password123',
}

# Connect to Switch
net_connect = ConnectHandler(**switch_device)

# Commands to execute using loop
commands = [
    "vlan 30",
    "name GUEST_GUEST_NET",
    "exit"
]

# Adding 10 ports using loop
for port in range(1, 11):
    commands.append(f"interface fastEthernet 0/{port}")
    commands.append("switchport mode access")
    commands.append("switchport access vlan 30")

output = net_connect.send_config_set(commands)
print("Automated Configuration Complete:")
print(output)
net_connect.disconnect()
```